



PES
UNIVERSITY

CELEBRATING 50 YEARS

COMPUTER NETWORKS

TEAM NETWORKS

Department of Computer Science and Engineering

COMPUTER NETWORKS

Link Layer and LAN

TEAM NETWORKS

Department of Computer Science and Engineering

COMPUTER NETWORKS

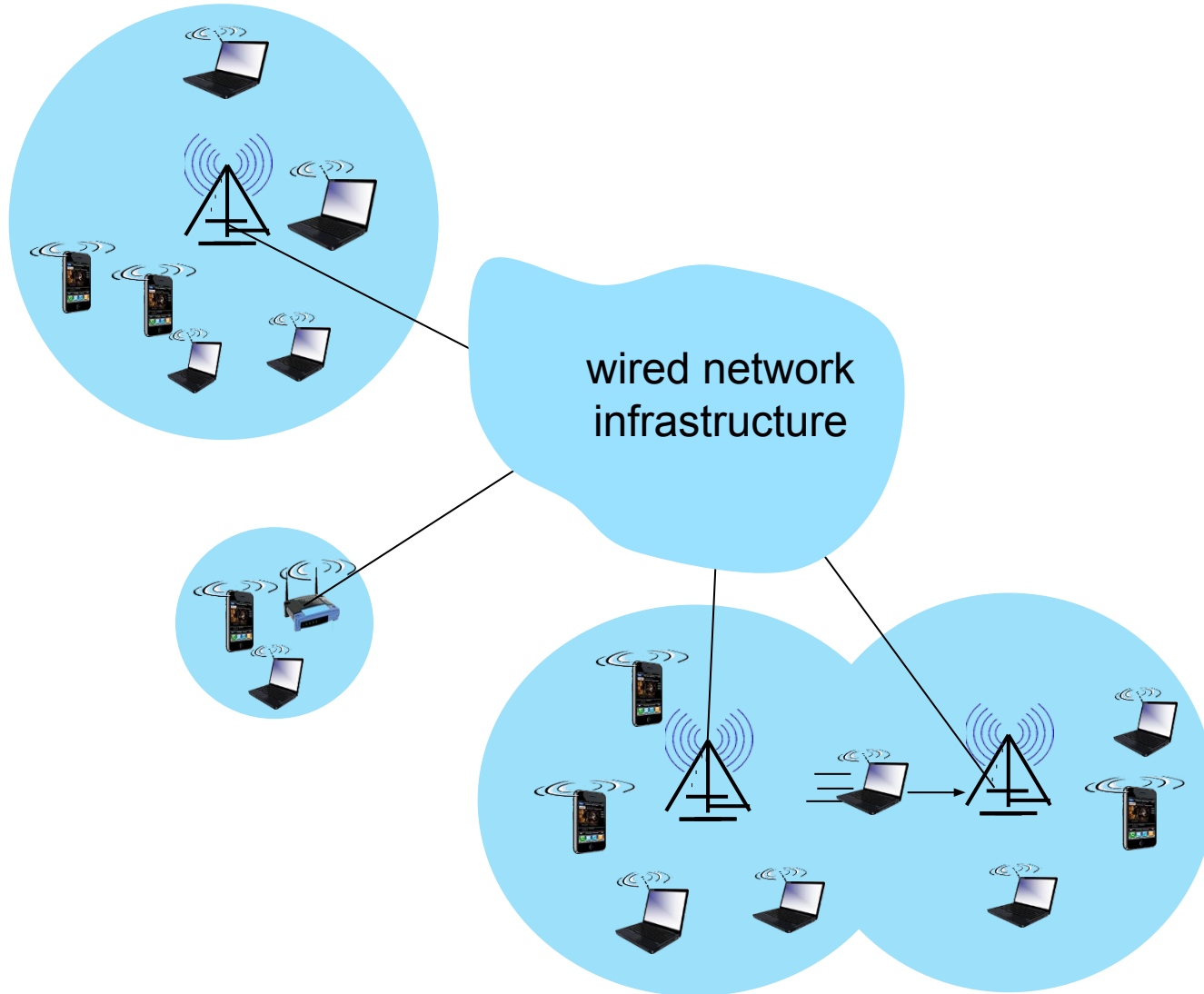
Unit – 4 Link Layer and LAN Roadmap



- Introduction
- Error detection, correction
- Multiple access protocols
- Switched LANs
 - addressing, ARP
 - Ethernet
 - switches
- A day in the life of a web request
- Physical layer
- Wireless LANs: IEEE 802.11

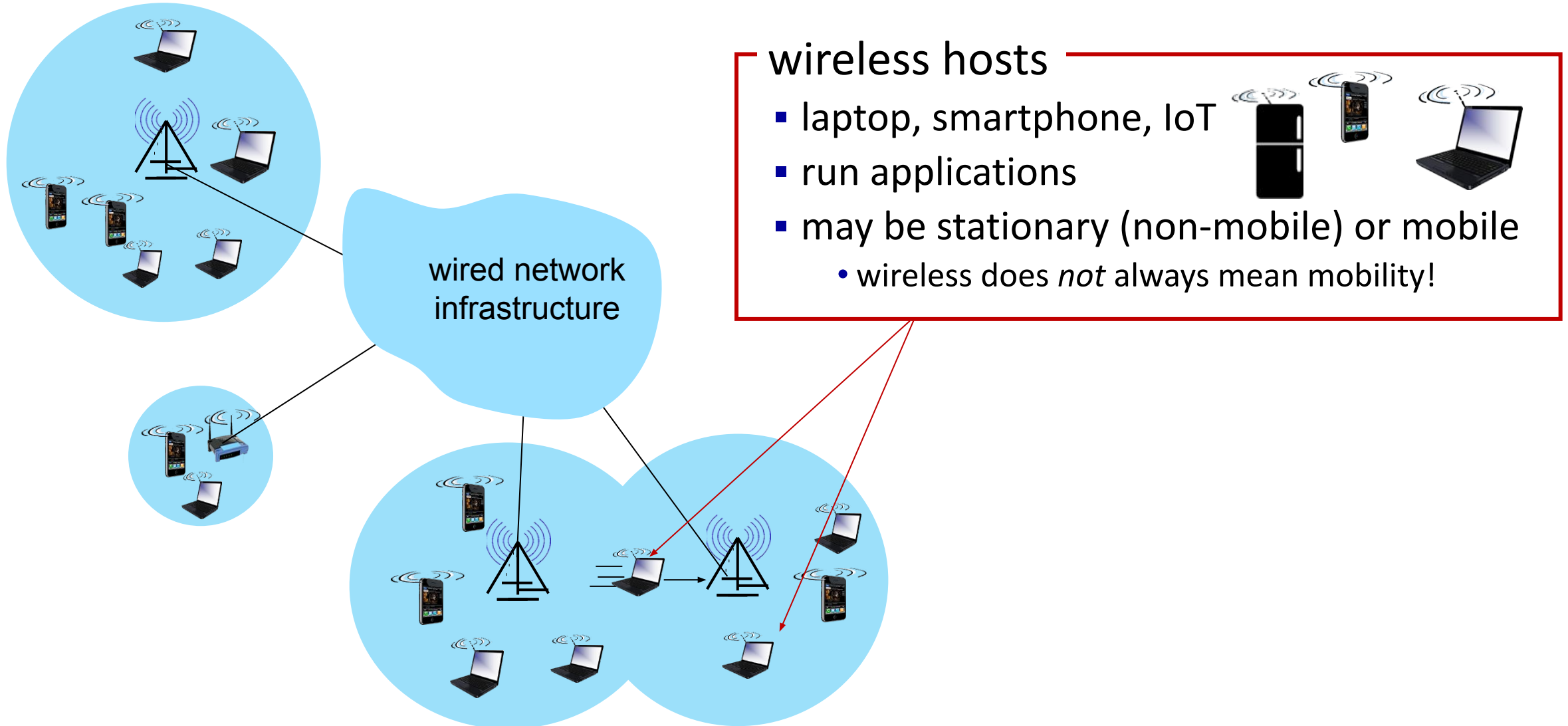
COMPUTER NETWORKS

Elements of a Wireless Network



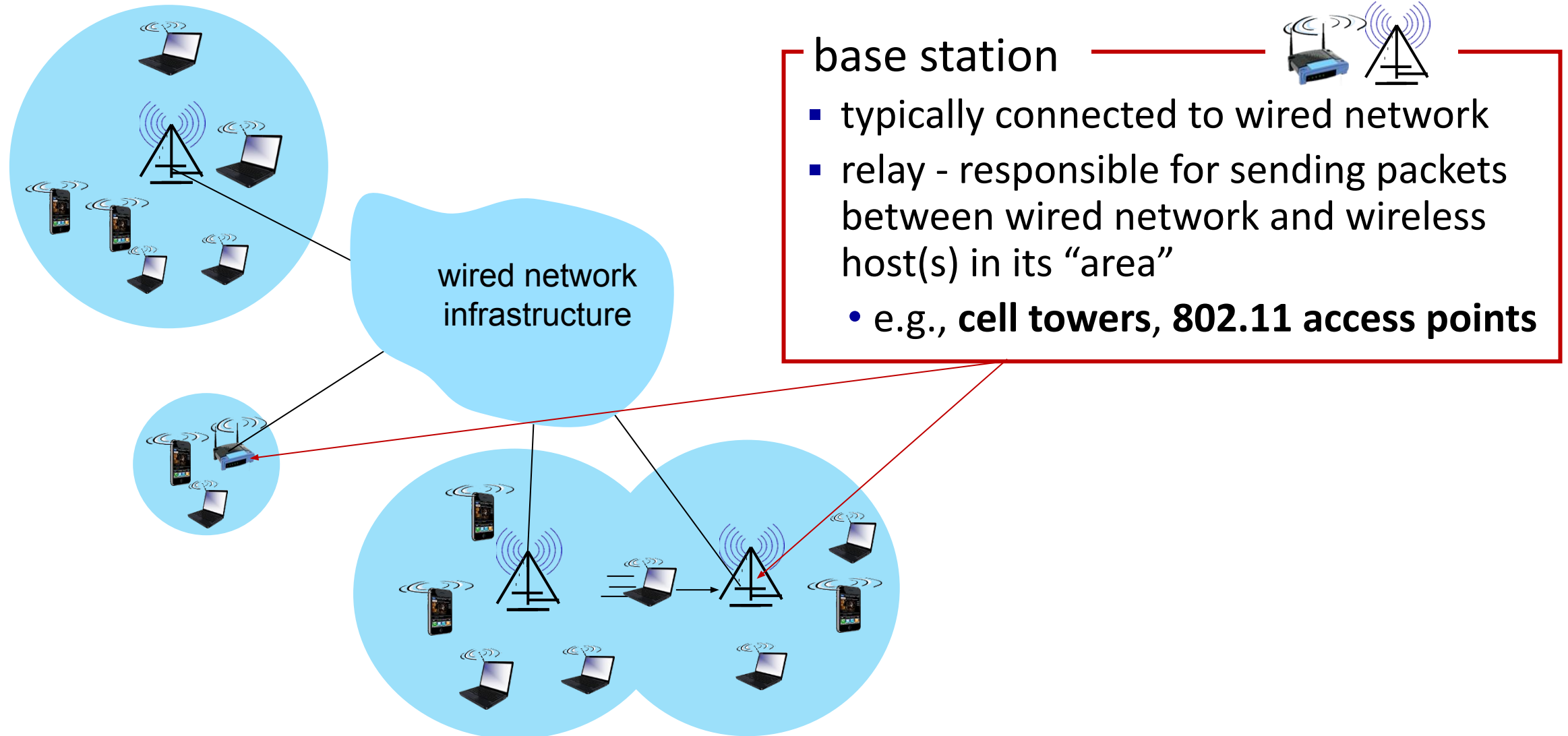
COMPUTER NETWORKS

Elements of a Wireless Network



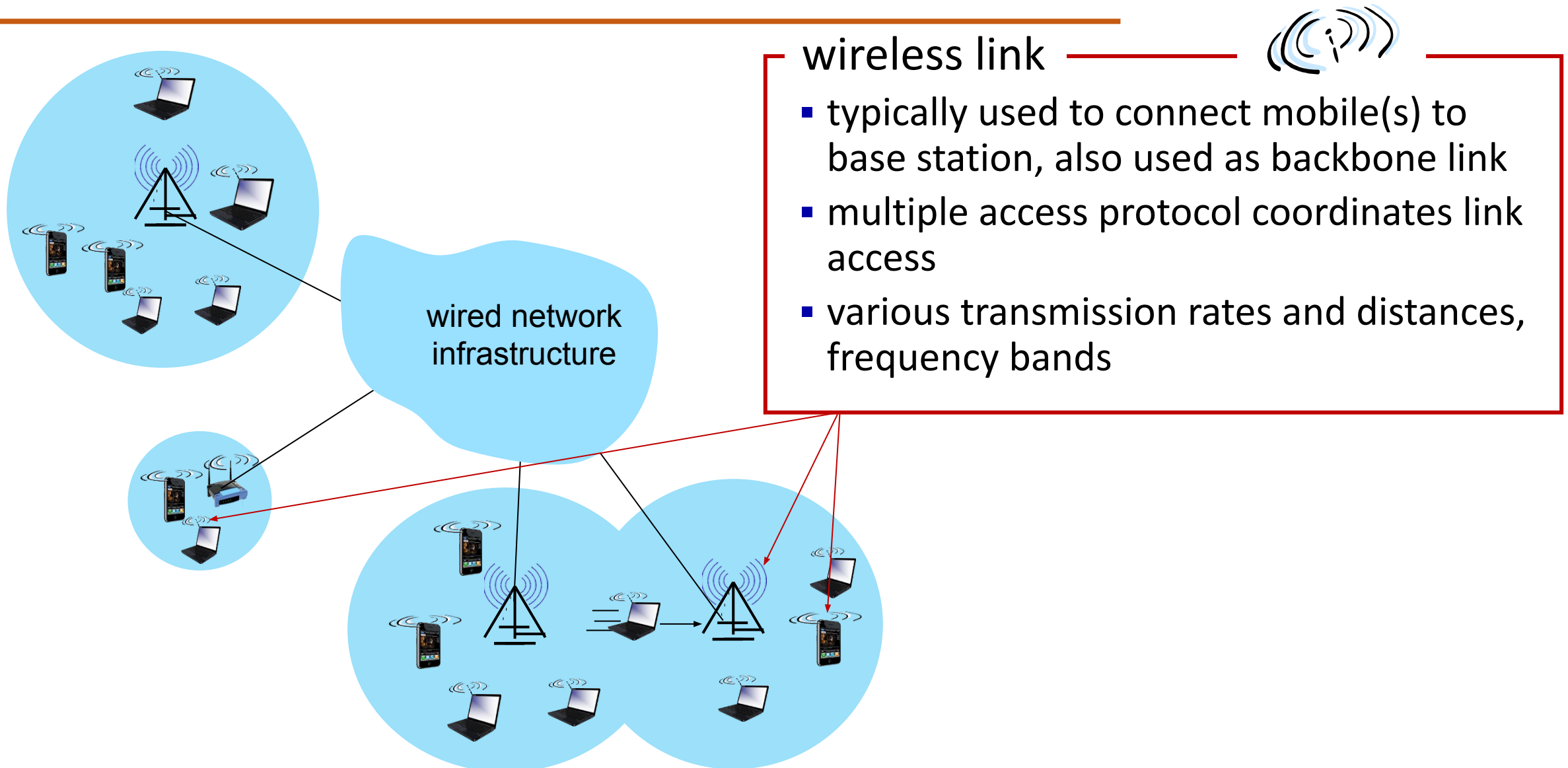
COMPUTER NETWORKS

Elements of a Wireless Network



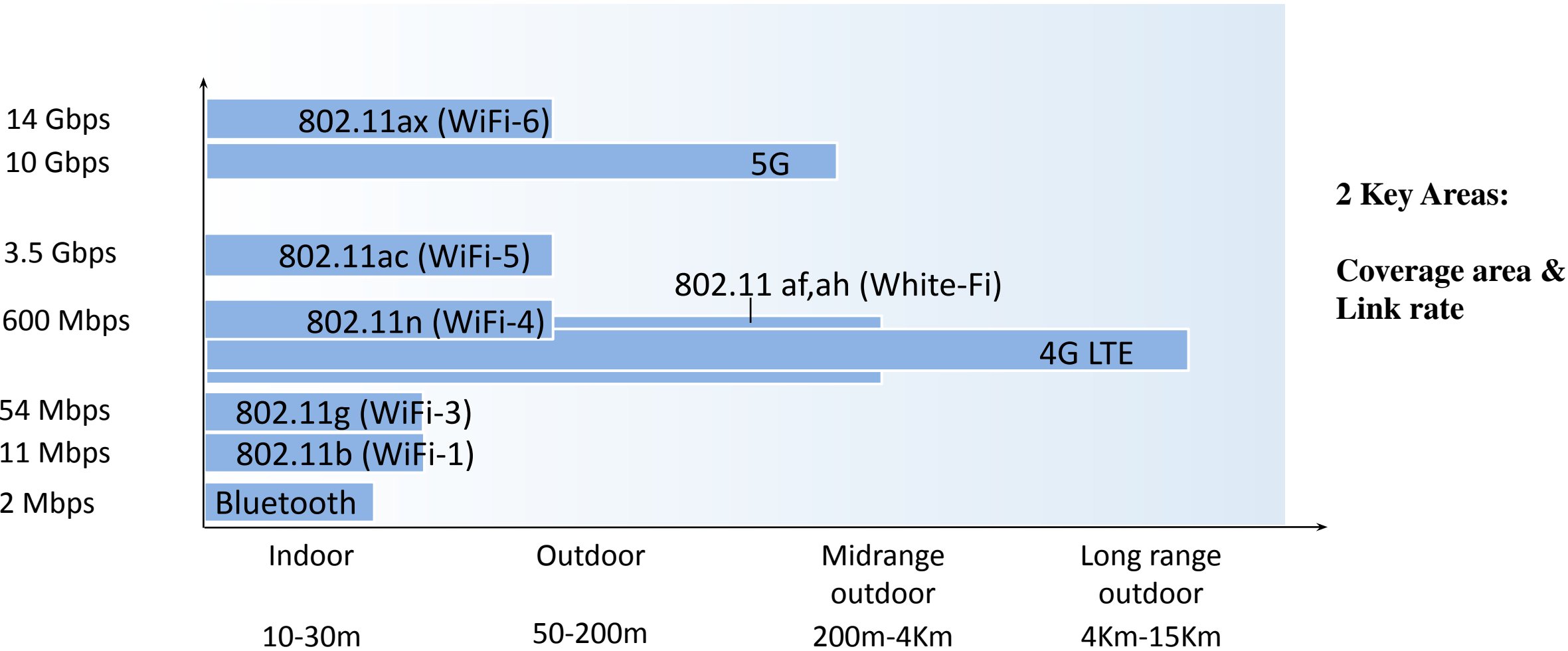
COMPUTER NETWORKS

Elements of a Wireless Network



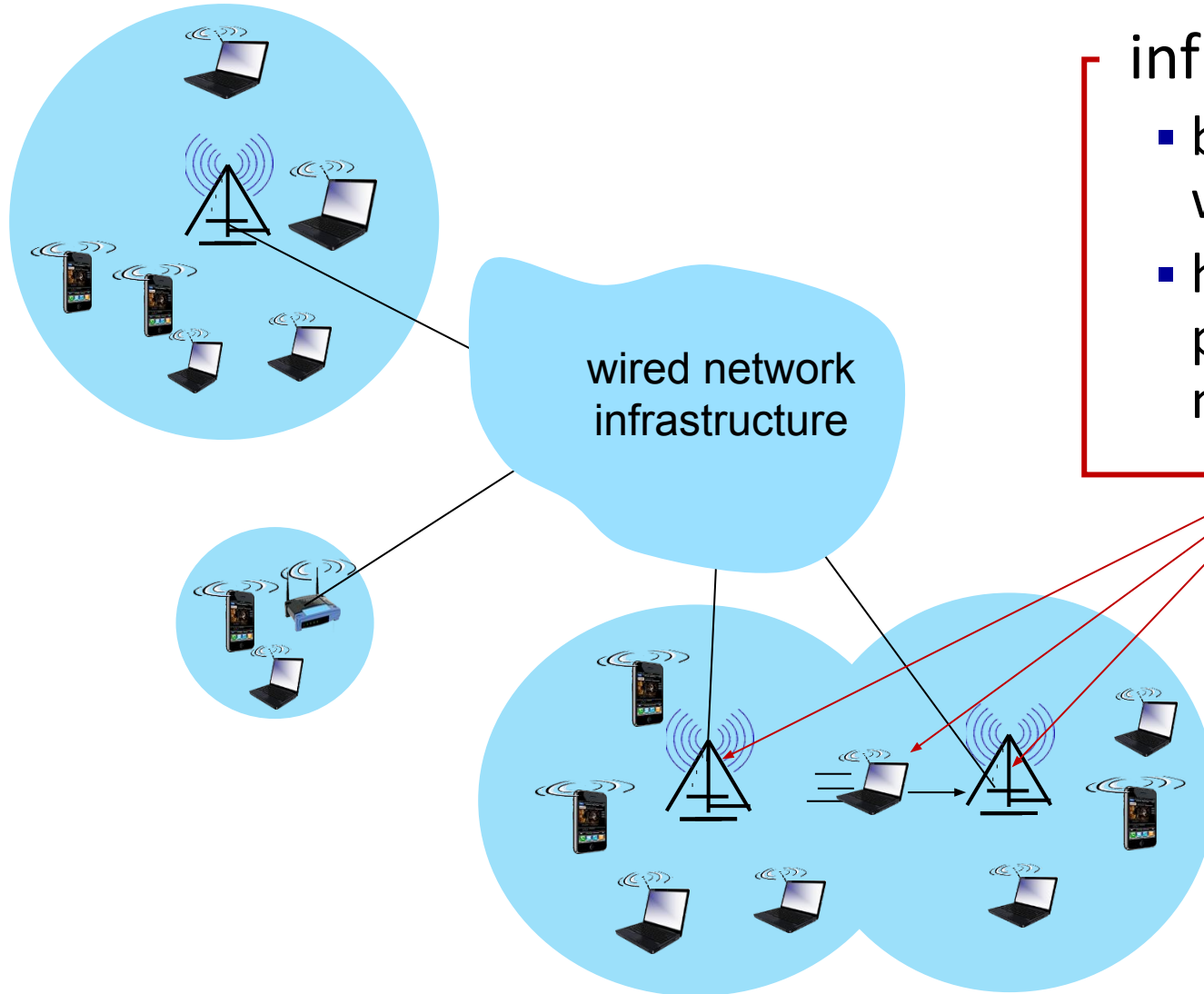
COMPUTER NETWORKS

Characteristics of Selected Wireless Links



COMPUTER NETWORKS

Elements of a Wireless Network

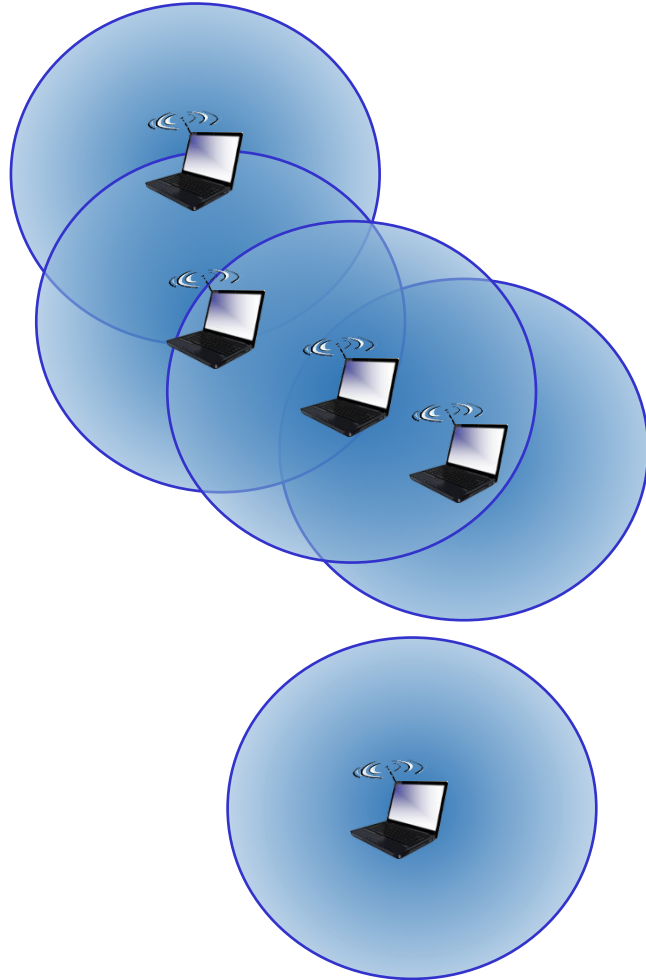


infrastructure mode

- base station connects mobiles into wired network
- handoff: mobile changes base station providing connection into wired network

COMPUTER NETWORKS

Elements of a Wireless Network



ad hoc mode

- no base stations
- nodes can only transmit to other nodes within link coverage
- nodes organize themselves into a network: routing, address assignment, DNS-like name translation, and more.

COMPUTER NETWORKS

Wireless network taxonomy

	single hop	multiple hops
infrastructure (e.g., APs)	host connects to base station (WiFi, cellular) which connects to larger Internet. Eg: 4G LTE	host may have to relay through several wireless nodes to connect to larger Internet: <i>mesh net</i>
infrastructure less	no base station, no connection to larger Internet. eg: Bluetooth, ad hoc nets	no base station, no connection to larger Internet. May have to relay to reach other. eg: MANET, VANET

COMPUTER NETWORKS

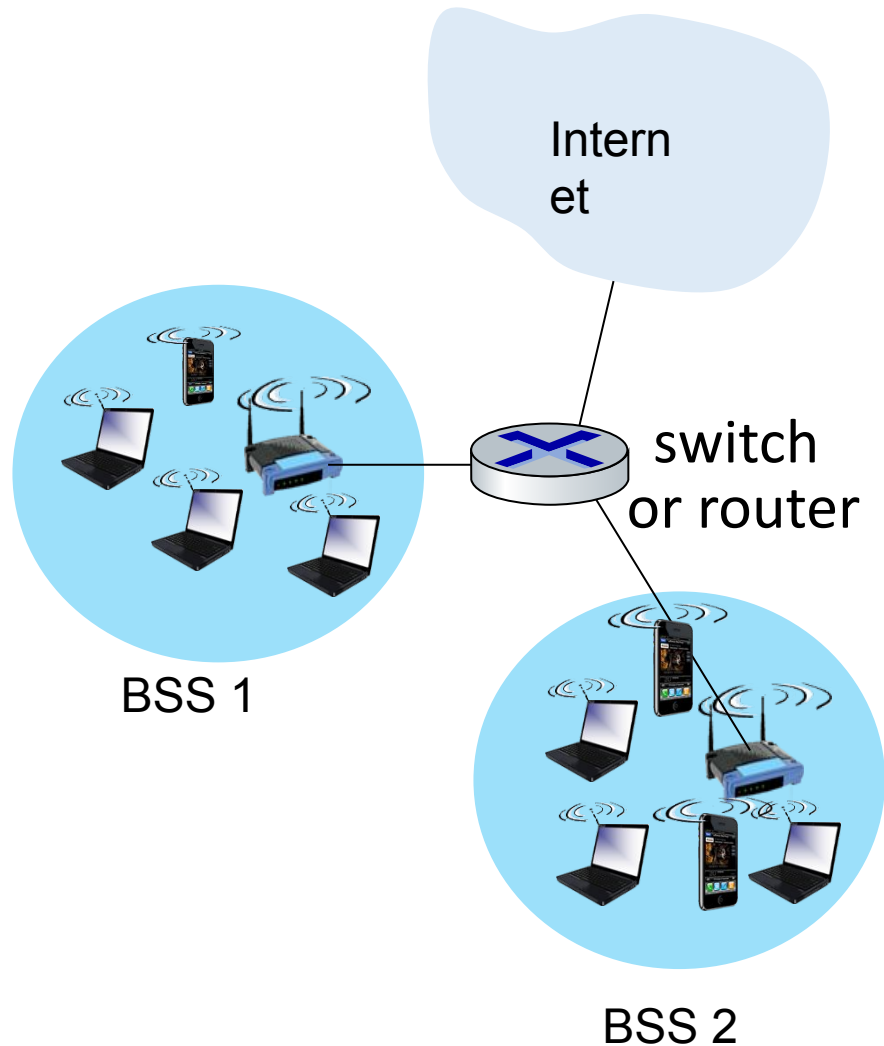
IEEE 802.11 Wireless LAN (WiFi)

IEEE 802.11 standard	Year	Max data rate	Range	Frequency
802.11b	1999	11 Mbps	30 m	2.4 Ghz
802.11g	2003	54 Mbps	30m	2.4 Ghz
802.11n (WiFi 4)	2009	600	70m	2.4, 5 Ghz
802.11ac (WiFi 5)	2013	3.47Gpbs	70m	5 Ghz
802.11ax (WiFi 6)	2020 (exp.)	14 Gbps	70m	2.4, 5 Ghz
802.11af	2014	35 – 560 Mbps	1 Km	unused TV bands (54-790 MHz)
802.11ah	2017	347Mbps	1 Km	900 Mhz

- all use CSMA/CA for multiple access, and have base-station and ad-hoc network versions

COMPUTER NETWORKS

The 802.11 LAN architecture



- wireless host communicates with base station
 - base station = access point (AP)
- Basic Service Set (BSS) (aka “cell”) in infrastructure mode contains:
 - wireless hosts
 - access point (AP): base station
 - ad hoc mode: hosts only

COMPUTER NETWORKS

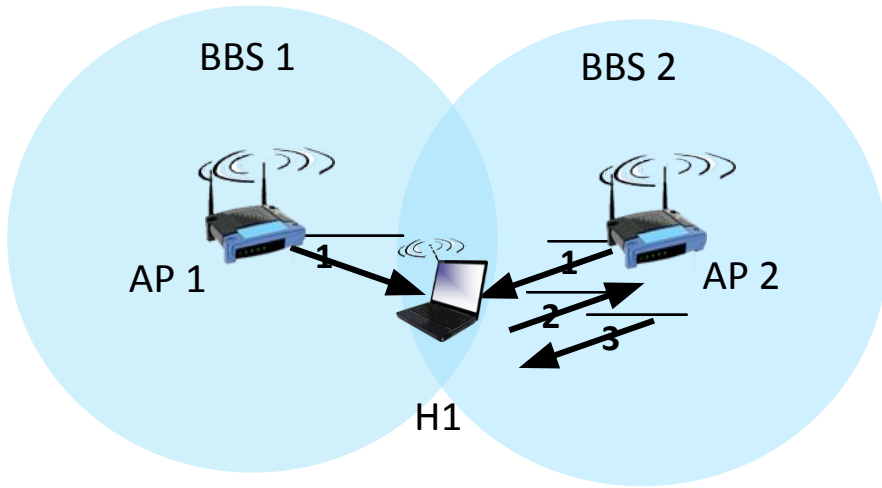
802.11: Channels, Association

- spectrum divided into channels at different frequencies
 - AP assigns **Service Set ID (SSID)**
 - AP admin chooses frequency for AP (**2.4 GHz to 2.4835 GHz**)
 - interference possible: channel can be same as that chosen by neighboring AP! – **WiFi Jungle**
- arriving host: must **associate** with an AP
 - scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
 - selects AP to associate with
 - then may perform authentication
 - then typically run DHCP to get IP address in AP's subnet



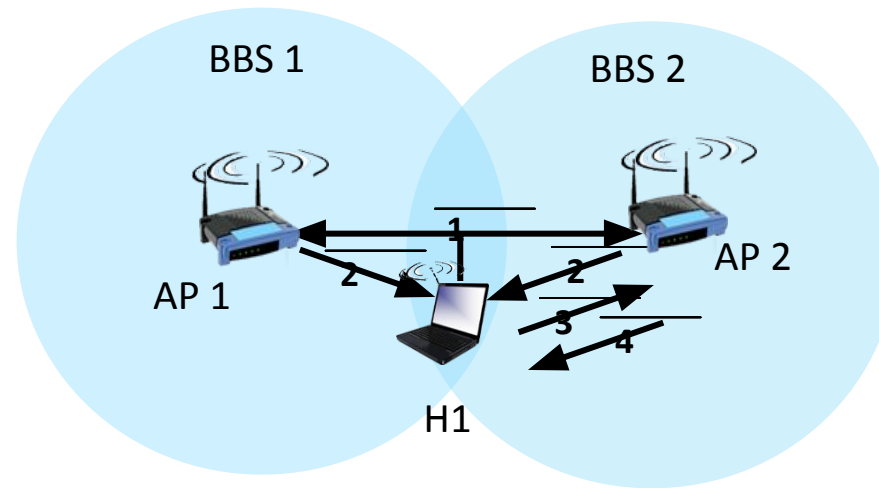
COMPUTER NETWORKS

802.11 active/passive scanning



passive scanning:

- (1) beacon frames sent from APs
- (2) association Request frame sent: H1 to selected AP
- (3) association Response frame sent from selected AP to H1



active scanning:

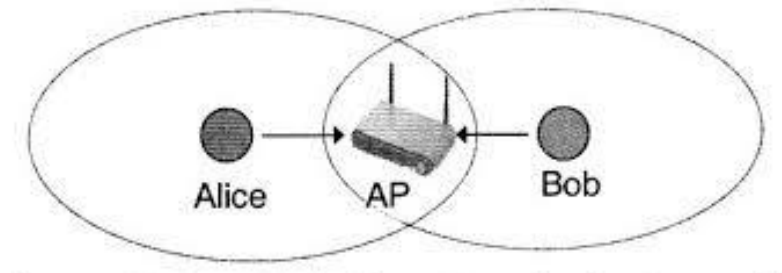
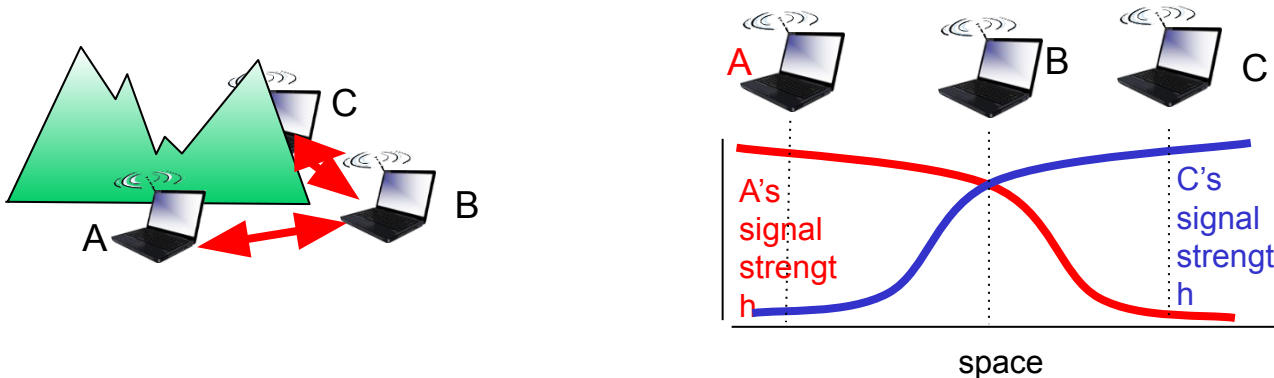
- (1) Probe Request frame broadcast from H1
- (2) Probe Response frames sent from APs
- (3) Association Request frame sent: H1 to selected AP
- (4) Association Response frame sent from selected AP to H1

COMPUTER NETWORKS

IEEE 802.11: multiple access

- avoid collisions: 2⁺ nodes transmitting at same time
- 802.11: CSMA - sense before transmitting
 - don't collide with detected ongoing transmission by another node
- 802.11: *no* collision detection!
 - difficult to sense collisions: high transmitting signal, weak received signal due to fading
 - can't sense all collisions in any case: hidden terminal, fading
 - goal: *avoid collisions*: CSMA/CollisionAvoidance

802.11's link-layer acknowledgment scheme!!!



COMPUTER NETWORKS

IEEE 802.11 MAC Protocol: CSMA/CA

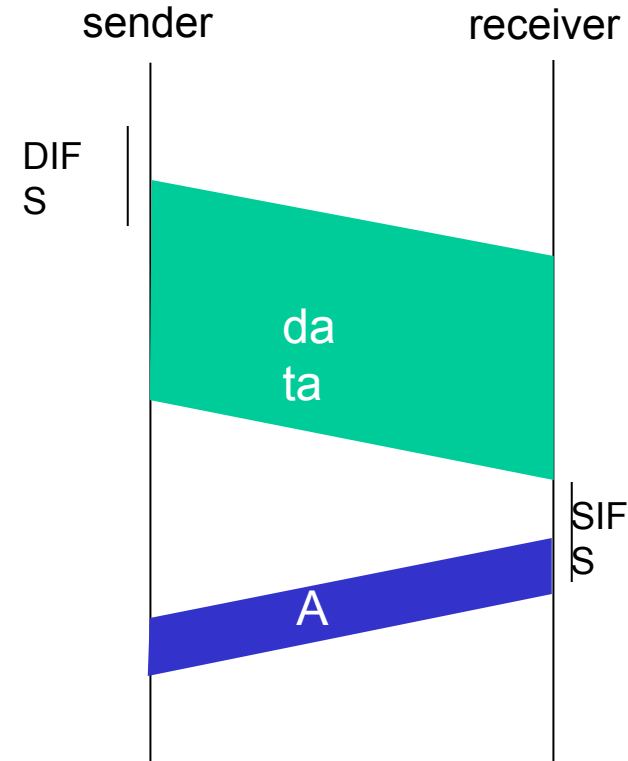
802.11 sender

Distributed Inter-frame Space (DIFS);

- 1 if sense channel idle for **DIFS** then
transmit entire frame (no CD)
- 2 if sense channel busy then
start random backoff time
timer counts down while channel idle
transmit when timer expires
if no ACK, increase random backoff interval, repeat 2

802.11 receiver

- if frame received OK
return ACK after **SIFS** (ACK needed due to hidden terminal problem)



Short Inter-frame Spacing (SIFS)

COMPUTER NETWORKS

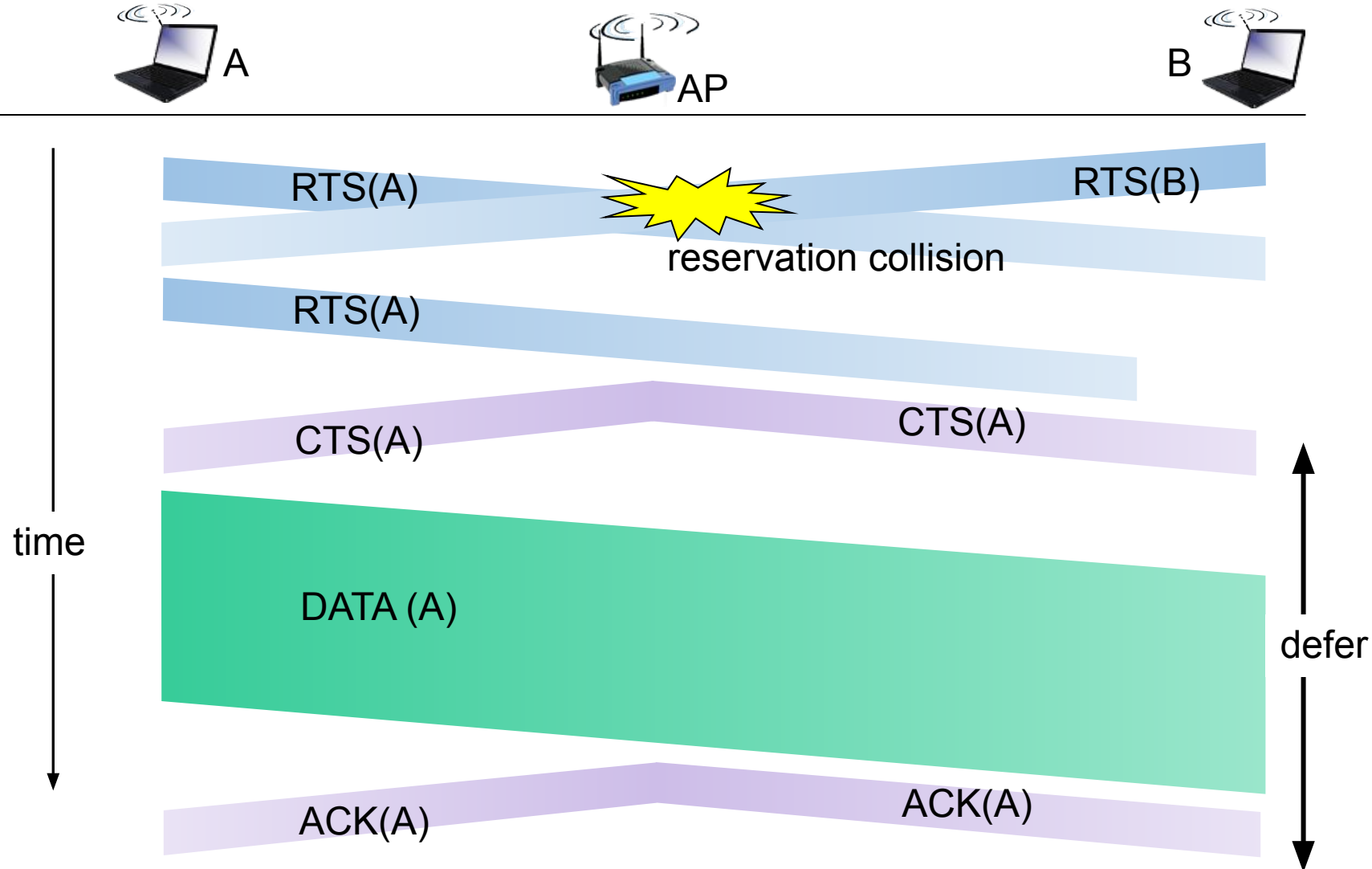
Avoiding Collisions (more)

idea: sender “reserves” channel use for data frames using small reservation packets

- sender first transmits *small* request-to-send (RTS) packet to BS using CSMA
 - RTSs may still collide with each other (but they’re short)
- BS broadcasts clear-to-send CTS in response to RTS
- CTS heard by all nodes
 - sender transmits data frame
 - other stations defer transmissions

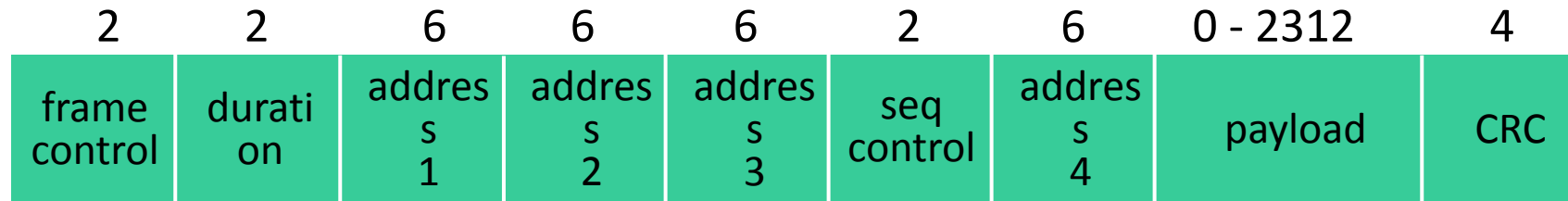
COMPUTER NETWORKS

Collision Avoidance: RTS-CTS exchange



COMPUTER NETWORKS

802.11 frame: addressing



Address 1: MAC address of wireless host or AP to receive this frame

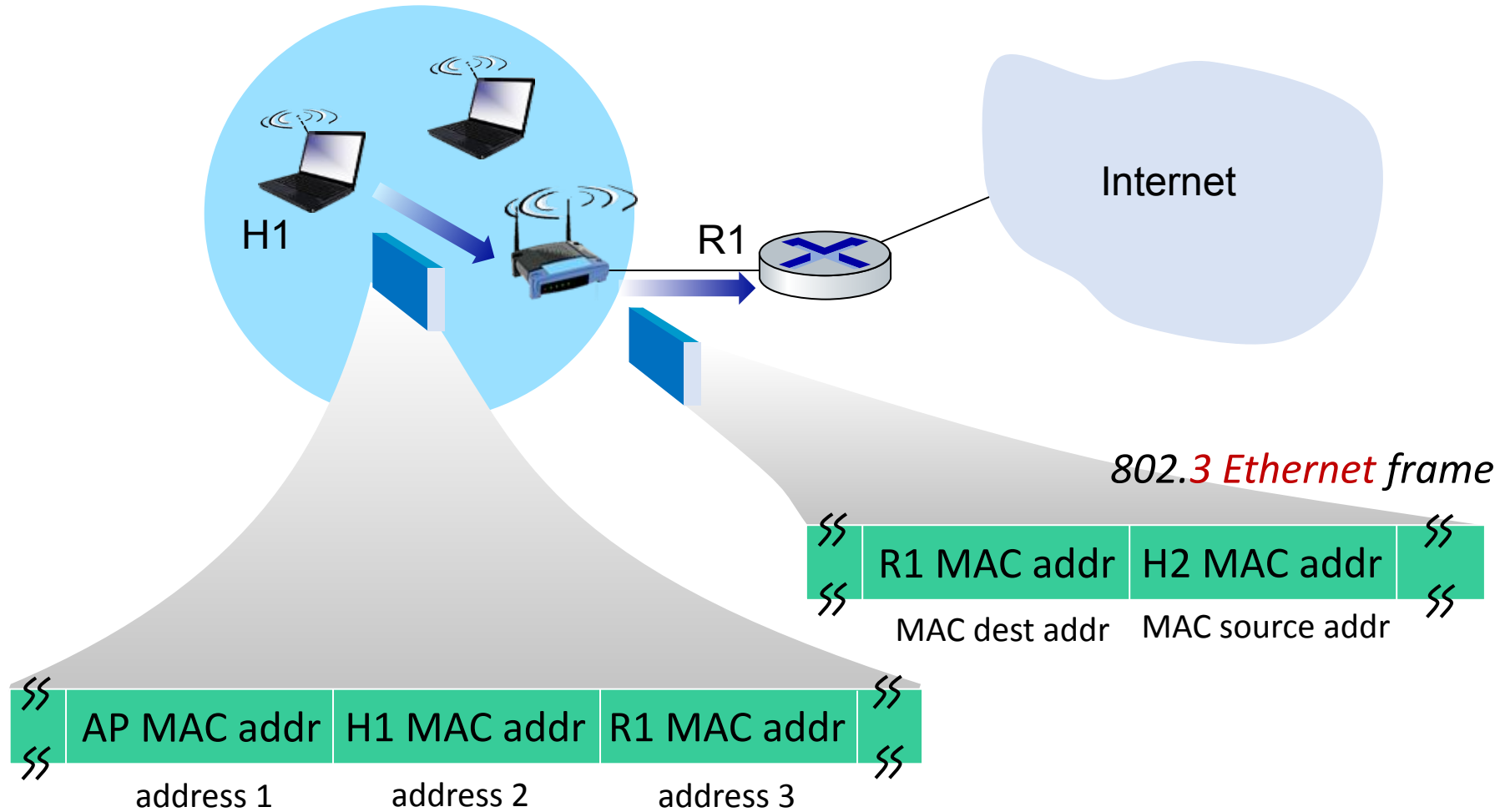
Address 2: MAC address of wireless host or AP transmitting this frame

Address 3: MAC address of router interface to which AP is attached

Address 4: used only in ad hoc mode

COMPUTER NETWORKS

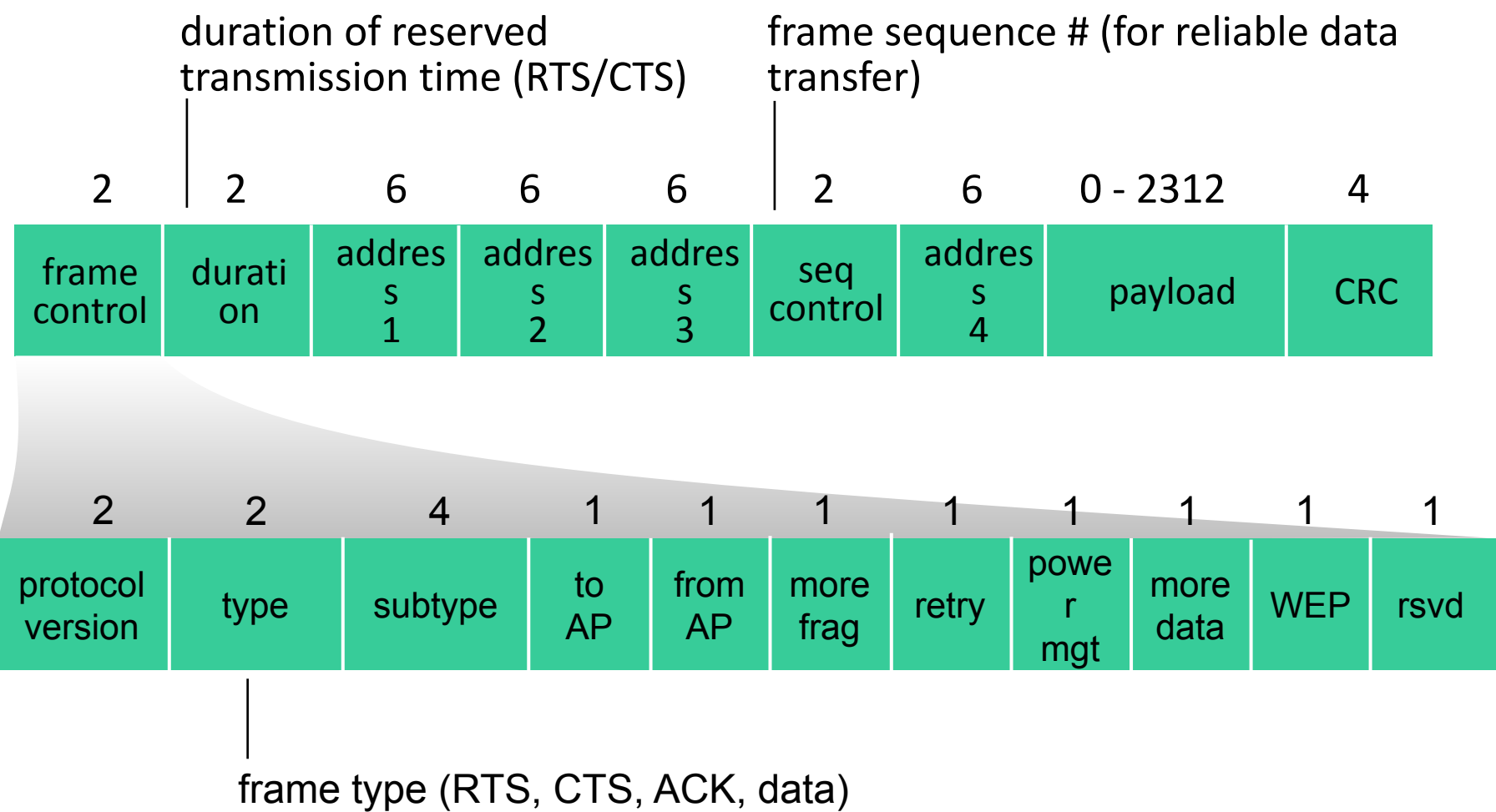
802.11 frame: addressing



802.11 WiFi frame

COMPUTER NETWORKS

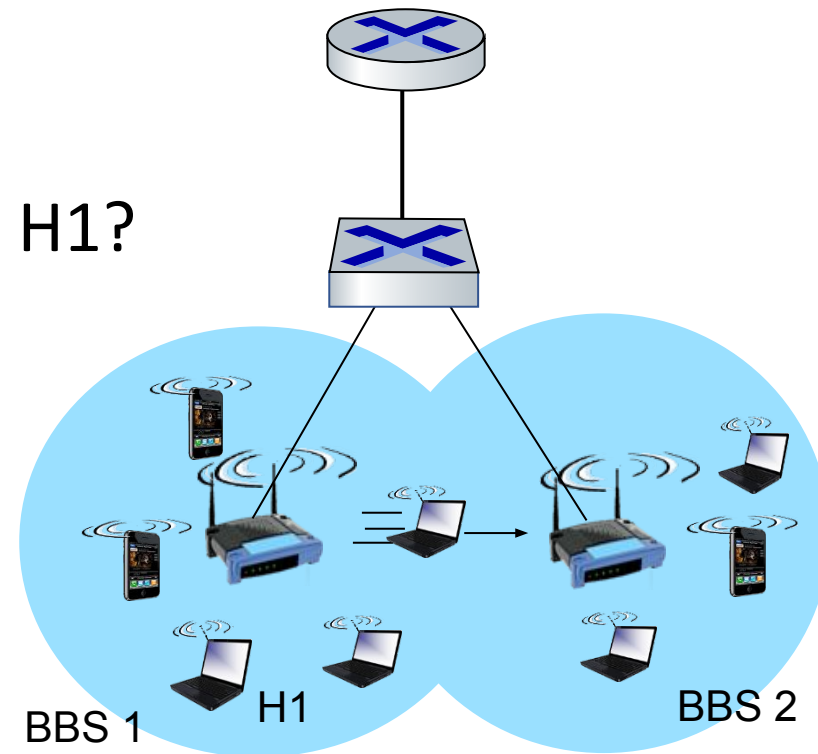
802.11 frame: addressing



COMPUTER NETWORKS

802.11: mobility within same subnet

- H1 remains in same IP subnet: IP address can remain same
- switch: which AP is associated with H1?
 - self-learning (Ch. 6): switch will see frame from H1 and “remember” which switch port can be used to reach H1





Thank You
For Your Attention



THANK YOU

TEAM NETWORKS

Department of Computer Science and Engineering